University of Sri Jayewardenepura

B.Sc. (General) Degree FirstYear

First Semester Terminal Course Unit Examination – May, 2017

ICT105 1.5 Computer Architecture

(Time: 1 ½ hours)

Answer **all**questions. Each question carries 25 marks. A calculator may be used.

Question 01 [25 marks]

a. The CPU time of a program is defined as the product of the CPI (cycles per instruction) for the processor on which it runs, the total number of instructions executed (I), and processor clock period (φ). Desribe the major factors which influence CPI, I and φ.

b. For a new architecture to be worth developing it must have a commercial lifespan of at least 10 years. What long-term factors must designers of a new architecture take into consideration during the design process?

c. Microprocessor core speeds increase at a rate of 40–60% per annum, compared with speed increases of 30% every ten years for DRAM devices. In the light of this increasing discrepancy between CPU and main memory speeds, what can architects, system designers and memory chip designers do to reduce the harmful effects of high memory latency in future computer systems?

Question 02 [25 marks]

The current trend toward ever faster processors and more modest increases in communications speeds requires that processors be able to tolerate ever longer delays between a request for data and its arrival. The choice of the unit data transferred between memory and cache, i.e the block size, can have a large impact on performance.

a. What factors determine the optimal block size for uniprocessor (a single processor) with blocking caches?

b. How are these factors affected by the introduction of a non-blocking cache?

c. How are these factors affected in the multiprocessor environment? What additional factors are introduced?

Question 03 [25 marks]

a. Calculate the effective CPI (taking into account both instruction execution and memory access) for the following unified instruction and data cache.

Instruction execution CPI = 1.0

Percent of instructions that are loads or stores = 33%

Cycles to access main memory = 100

Cycles to access cache = 2

Miss rate for cache 2.0%

b. What is the total memory access rate (memory access per cycle)?

c. How many cycles per instruction are taken for handling misses?

d. How many cycles per instruction are taken for handling hits?

e. What is the effective CPI for the System with the cache?

Question 04 [25 marks]

a. Describe the graphics pipeline of a Graphical Processing Unit (GPU)?

b. How does marker-based motion capture system create human's skeleton in real time and describe how GPU contributes to the process based on your views?